

GEOFENCING: Technology's Virtual 'Circle of Trust'



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*The incredibly
interesting geofencing
software can be put
to use in many kinds
of applications by
almost anyone*

Geofencing, as the name suggests, is a location-based technology that creates a virtual boundary or fence. It is a feature in a software program that uses the global positioning system (GPS) or radio frequency identification (RFID) to define a virtual geographical boundary, establishing a radius of interest that can trigger an action in a geo-enabled phone or any other portable electronic device. It is a very useful technology to define a virtual boundary around a real-world geographical area, to generate alerts based on the defined coordinates of a geographic area.

Geo-location technology allows mobile apps to do incredible things, and geofencing is the next step in this ground-breaking innovation. One of the first users of geofence were farmers who would equip a herd of cattle with GPS units. When the herd of cattle moved out of the geofence set by the farmer, the farmer would receive an alert.

Today, geofencing has moved from its expensive commercial use to the realm of consumer application and is widely being incorporated by developers and innovators into their hardware (smartphones, tablets, etc.). There are a large number of applications that offer the benefits of geofencing to people like you and me.

As per a recent report from ABI Research, the market for geofencing tools currently stands at about \$300 million. With low-cost developer tools becoming accessible, geofencing is coming to the

forefront and moving beyond conventional location-based applications to form the backbone of a host of new applications and services.

In the past, geofencing tools have been tough to develop, costly to licence and known to have limited application areas. With location features now being present in almost all handset types and a host of geofencing tools available from a number of major players, geofencing is expected to soon become an integral part of our daily lives.

How it works

Geofencing allows us to set up automatic alert triggers when a device enters or exits the boundaries defined by the administrator. For example, if the geofence is a geographic virtual boundary surrounding a user's house, an e-mail or text message is automatically triggered and sent to the user by a geofence-enabled app on the phone when his child with the cell phone enters this area.

It is possible to establish a geofence area around a neighbourhood and have alerts sent out on specified days. For example, alerts could be sent out by a collection company on days fixed for garbage collection as per the schedule entered in geofence-enabled app.

By incorporating Google Earth, administrators can define boundaries superimposed on top of a satellite view of a specific geographical area. Or, they can do this by longitude and latitude, or through user-created and web-based maps.

Geofence virtual barriers can be either active or passive. Active geofences require the end user to turn on location services and open a mobile app. Passive geofences are 'always on,' rely on Wi-Fi and cellular data instead of GPS or RFID, and quietly work in the background.

Geofencing software

A good geofencing software is one that provides a host of information

Fig. 1: Geolocation-based
marketing methods
(Source: www.appboy.com)

